

Group Project 3: Intelligent Exploration of Educational Videos

Degree Program: Master of Science in Computer Science

ECTS Credits: 12 ECTS per student

Duration: 2 semesters

Group Size: 3 students

Workload: approx. 300-360 hours per student

1. Project Background and Motivation

Educational videos are a cornerstone of modern learning, but navigating hour-long lectures and understanding connections across video collections remains challenging. While platforms like the TIB AV-Portal host thousands of scientific videos, users often struggle to efficiently explore content, identify relevant segments, and compare multiple videos.

This project develops an intelligent system for exploring educational video collections through automatic segmentation, multi-level summarization, and interactive visualization. You will implement state-of-the-art video chaptering techniques, create meaningful summaries at both individual and collection levels, and design intuitive visual interfaces for exploration.

Scientific Foundation:

- **Data Source:** TIB AV-Portal (<https://av.tib.eu/>) - scientific video repository
- **Video Chaptering:** Ventura et al. (2025), "Chapter-Llama: Efficient Chaptering in Hour-Long Videos with LLMs", CVPR 2025, https://openaccess.thecvf.com/content/CVPR2025/html/Ventura_Chapter-Llama_Efficient_Chaptering_in_Hour-Long_Videos_with_LLMs_CVPR_2025_paper.html
- **Visualization:** Afzal et al. (2023), "Visualization and Visual Analytics Approaches for Image and Video Datasets: A Survey", <https://dl.acm.org/doi/full/10.1145/3576935>

2. Learning Objectives

During the project, students will:

- Implement and extend state-of-the-art video understanding techniques
- Design systems for multimodal content analysis using open-source language models
- Create multi-level summarization systems for video collections
- Develop interactive visualization interfaces for complex data
- Process and analyze long-form video content efficiently
- Apply visual analytics principles to educational technology
- Evaluate user-centered systems with domain experts

3. Project Task

Develop an intelligent exploration system for educational video collections that enables efficient navigation, understanding, and comparison of video content through automatic segmentation, summarization, and interactive visualization.

3.1 Sub-Task 1: Video Segmentation and Chaptering (Student A)

Task Description: Implement an automatic video segmentation system that partitions educational videos into semantic chapters with descriptive titles, based on the Chapter-Llama framework.

Specific Requirements:

Core Segmentation Pipeline:

- **Multimodal Feature Extraction:**
 - Extract speech transcripts using ASR (Whisper or similar)
 - Generate frame captions using vision-language models
 - Implement timestamp alignment for all modalities
- **Frame Selection Strategy:**
 - Implement speech-guided frame sampling
 - Balance computational efficiency with information coverage
 - Handle topic changes and visual transitions
- **LLM-Based Chaptering:**
 - Feed transcripts and captions with timestamps to LLM
 - Generate chapter/topic boundary predictions
 - Create descriptive, informative chapter titles
 - Handle videos up to 1-2 hours in length

Model Implementation:

- **Architecture:**
 - Use large-context LLMs (e.g. Llama 3.1 with extended context)
 - Implement prompt engineering for chaptering task
 - Design output format (timestamps + titles)
- **Optimization:**
 - Efficient processing of long-context inputs
 - Batching strategies for multiple videos
 - Quality vs. speed trade-offs

Evaluation:

- **Metrics:**
 - Boundary detection accuracy (F1, precision, recall)
 - Title quality (BLEU, ROUGE, human evaluation)
 - Temporal consistency
- **Comparison:**
 - Baseline methods (equal-length segments, transcript-only)
 - Ablation studies (with/without frame captions)
 - User preference studies

Technologies (Recommendations):

- ASR: OpenAI Whisper
- Vision-Language: GLM-4.5, BLIP-2, LLaVA
- LLMs: Open source LLMs like Llama 3.1, Mistral, Mixtral
- Video Processing: FFmpeg, OpenCV, PySceneDetect

Deliverables:

- Video segmentation pipeline implementation,
- Chapter boundary and title generation module

- Evaluation framework and results
- Documentation of approach and findings, including API for segmentation service

3.2 Sub-Task 2: Multi-Level Video Summarization (Student B)

Task Description: Develop a comprehensive summarization system that creates individual video summaries, identifies commonalities across video collections, and highlights differences between related videos.

Specific Requirements:

Individual Video Summarization:

- **Content-Based Summarization:**
 - Generate concise summaries (3 levels: short/medium/long)
 - Extract key concepts and topics
 - Identify main arguments and conclusions
 - Create learning objectives from content
- **Visual-Textual Integration:**
 - Combine transcript analysis with visual information
 - Identify important visual elements (diagrams, formulas, demonstrations)
 - Create multimodal summaries referencing both speech and visuals
- **Chapter-Aware Summarization:**
 - Use segmentation from Sub-Task 1
 - Generate chapter-level summaries
 - Create hierarchical summary structure

Collection-Level Analysis:

- **Commonality Detection:**
 - Identify shared topics across video collection
 - Extract common concepts and themes
 - Find repeated patterns and structures
 - Generate collection overview summaries
- **Difference Highlighting:**
 - Compare multiple videos on same topic
 - Identify unique content in each video
 - Highlight different perspectives and approaches
 - Create comparative summaries
- **Relationship Mapping:**
 - Detect prerequisite relationships
 - Identify progressive difficulty levels
 - Map conceptual dependencies

LLM-Based Implementation:

- **Summarization Strategies:**
 - Extractive and abstractive summarization
 - Multi-document summarization for collections
 - Query-focused summarization (optional)
- **Prompting Techniques:**
 - Chain-of-thought for complex summaries
 - Few-shot learning with examples

- Structured output generation (JSON)

Evaluation:

- **Summary Quality:**
 - ROUGE scores for extractive quality
 - Human evaluation (informativeness, coherence)
 - Coverage of key topics
- **Collection Analysis:**
 - Accuracy of commonality detection
 - Usefulness of difference highlights
 - Expert feedback from educators

Technologies (Recommendations):

- LLMs: Open-source models like Llama 3.1, Mistral, Mixtral
- NLP: Transformers, spaCy, NLTK
- Embeddings: Sentence-BERT, MPNet, GTE
- Topic Modeling: BERTopic, LDA (optional)

Deliverables:

- Individual video summarization module
- Collection-level analysis system
- Commonality and difference detection algorithms
- Summary generation API
- Evaluation report with quality metrics
- Example summaries for different video types

3.3 Sub-Task 3: Interactive Visualization Interface (Student C)

Task Description: Design and implement an intuitive, interactive visualization system that presents segmentation and summarization results in a way that facilitates efficient video exploration and comparison.

Specific Requirements:

Video-Level Visualizations:

- **Timeline View:**
 - Interactive video timeline with chapter markers
 - Chapter titles and durations
 - Hover previews showing chapter summaries
 - Clickable segments for direct navigation
- **Summary Display:**
 - Hierarchical summary presentation (overview → detailed)
 - Key concept highlighting
 - Visual content preview (screenshots, diagrams)
 - Expandable/collapsible sections
- **Content Overview:**
 - Topic tag cloud or visualization
 - Concept relationship graphs
 - Learning objective display
 - Video metadata and statistics

Collection-Level Visualizations:

- **Similarity and Comparison Views, e.g.**
 - Network graphs showing video relationships
 - Matrix visualizations for pairwise comparisons
 - Clustering visualizations by topic
 - Timeline showing video release/creation dates
- **Commonality Visualization, e.g.**
 - Venn diagrams for shared topics
 - Heat maps showing topic coverage
 - Hierarchical clustering of common themes
 - Interactive exploration of shared concepts
- **Difference Highlighting, e.g.**
 - Side-by-side comparison interface
 - Synchronized timeline comparison
 - Unique content markers
 - Differential topic analysis

Interactive Features:

- **Navigation:**
 - Search within and across videos
 - Jump to specific chapters or topics
 - Filter by metadata (speaker, date, topic)
 - Bookmark and create playlists
- **Exploration Support:**
 - Drill-down from overview to detail
 - Linked views (select chapter → show summary)
 - Brushing and linking across visualizations
 - Export and share functionality
- **User Preferences:**
 - Adjustable summary detail level
 - Customizable visualization layouts
 - Save personal notes and annotations
 - Progress tracking

Technical Implementation:

- **Frontend Architecture:**
 - Responsive web application
 - Component-based design
 - Real-time updates
 - Mobile-friendly interface
- **Video Player Integration:**
 - Custom controls with chapter navigation
 - Synchronized multi-video playback (for comparison)
 - Speed control and subtitles
 - Screenshot capture

Evaluation:

- **Usability Testing:**
 - Task completion rates

- Time-on-task measurements
- User satisfaction surveys
- Think-aloud protocols
- **Expert Review:**
 - Feedback from educators/students
 - Assessment of pedagogical value
 - Suggestions for improvement

Technologies (Recommendations):

- Frontend: React, Vue.js, Svelte
- Visualization: D3.js, Plotly, ECharts, Recharts
- Video Player: Video.js, Plyr
- UI Components: Material-UI, Ant Design, shadcn/ui
- State Management: Redux, Zustand, Pinia

Deliverables:

- Complete web-based visualization interface
- Multiple coordinated views for exploration
- Interactive comparison tools
- User manual and documentation
- Usability study report
- Demo video showcasing features

4. Project Phases and Milestones

Phase 1: Research and Setup

- Literature review (video segmentation, summarization, visualization)
- Access and explore TIB AV-Portal videos
- Set up development environment
- Design system architecture
- **Milestone M1:** Design document and video dataset selection

Phase 2: Core Implementation

- Implement segmentation pipeline
- Develop summarization modules
- Create basic visualization prototypes
- Initial integration
- **Milestone M2:** Functional prototypes of all components

Phase 3: Integration and Testing

- Connect all components
- End-to-end testing
- Bug fixes and refinements
- Initial user feedback
- **Milestone M3:** Integrated system demonstration + interim presentation

Phase 4: Enhancement

- Improve segmentation quality
- Enhance summarization depth
- Refine visualizations based on feedback
- Add advanced features
- **Milestone M4:** Production-ready enhanced system

Phase 5: Evaluation

- Conduct usability studies
- Quantitative evaluation
- Expert interviews
- Comparative analysis
- **Milestone M5:** Complete evaluation report

Phase 6: Finalization

- Final refinements
- Complete documentation
- Prepare presentation and demo
- Write project report
- **Milestone M6:** Final submission and presentation

5. Technical Requirements

5.1 Development Environment

- **Languages:** Python 3.10+, JavaScript/TypeScript
- **Version Control:** Git (GitHub/GitLab)
- **CI/CD:** GitHub Actions or GitLab CI
- **Code Quality:** pytest, black, mypy, ESLint

5.2 Hardware and API Requirements

- **Computing Resources:**
 - **Local development:** Standard development machine
 - **Model inference:** Access to university GPU resources (if available)
- **Storage:** Sufficient space for video processing

5.3 Data Sources

- **TIB AV-Portal:** Scientific lecture videos
- **Test Dataset:** 20-50 educational videos from various domains
- **Evaluation Data:** Ground truth chapters and summaries (create or collect)
- Test data and evaluation should focus on domains of available expertise (CS, data science)

6. Deliverables

Technical Deliverables

1. **Video Segmentation System:** Working implementation
2. **Summarization System:** Individual and collection-level
3. **Visualization Interface:** Interactive web application
4. **Source Code:** Complete, documented, tested
5. **API Documentation:** RESTful API specifications

6. **Deployment Package:** Docker containers and setup

Scientific Deliverables

1. **Project Report (~10 pages):**
 - Introduction and motivation
 - Related work
 - Methodology (detailed per sub-task)
 - Implementation details
 - Evaluation results
 - User study findings
 - Discussion and lessons learned
 - Conclusion and future work
2. **Final Presentation (20 min + 10 min Q&A):**
 - Live demo of complete system
 - Key findings and insights
 - Evaluation highlights

7. Assessment Criteria

- **Technical Implementation**
- **Scientific Work**
- **Teamwork**
- **Individual Performance**

8. Evaluation Focus Areas

8.1 Segmentation Quality

- Accuracy of chapter boundaries
- Relevance of chapter titles
- Processing efficiency

8.2 Summarization Quality

- Individual summary accuracy and coverage
- Collection-level analysis usefulness
- Commonality and difference detection accuracy

8.3 Visualization Effectiveness

- User satisfaction and usability
- Task completion efficiency
- Insight discovery support
- Interface intuitiveness

8.4 User Study

- **Participants:** Students, educators, researchers (4-7 people)
- **Tasks:**
 - Find specific information in videos
 - Compare multiple related videos
 - Explore video collection by topic
- **Metrics:**

- Task success rate
- Time-on-task
- Subjective satisfaction
- Qualitative feedback

9. Recommended Resources

Papers and Literature

- **Core Papers:** Ventura et al. (2025), Afzal et al. (2023)
- **Video Understanding:** Video summarization surveys, action segmentation
- **LLMs for Video:** VideoLLaMA, Video-ChatGPT, related works
- **Visualization:** Interactive video visualization, visual analytics

Tools and Datasets

- **Video Processing:** FFmpeg, OpenCV, PySceneDetect
- **ASR:** Whisper (open-source), Mozilla DeepSpeech
- **Vision-Language:** e.g. BLIP-2, LLaVA
- **LLMs:** Open source LLMs Llama 3.1
- **Visualization:** D3.js, Plotly, Video.js

10. Optional Extensions

- **Multi-Language Support:** Handle non-English videos
- **Accessibility Features:** Generate audio descriptions, enhanced subtitles
- **Personalized Learning Paths:** Recommend viewing order based on prerequisites
- **Quiz Generation:** Automatic question generation from content
- **Integration with LMS:** Export to Moodle, Canvas
- **Mobile App:** Native mobile interface
- **Real-Time Processing:** Live video segmentation during streaming

11. Notes on Individual Work

Each student leads his/her sub-task:

- **Student A:** Video segmentation and chaptering
- **Student B:** Multi-level summarization
- **Student C:** Interactive visualization

All students must:

- Collaborate on system architecture and integration, review each other's code
- Contribute to evaluation design
- Co-author project report
- Document individual contributions

Critical Integration Points:

- Segmentation output format for visualization
- Summary data structure for display
- Common API for accessing video data
- Unified user experience

Good luck with your project!